

BESTROUTE
Intelligent Public Transportation
Optimization System

Backend API Testing Report

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1. Introduction

BestRoute is an intelligent multimodal public transportation optimization platform developed by Team CodeCrushers under Competition Theme 05. The system is designed to support passengers in planning, comparing, and optimizing journeys involving multiple transportation modes.

The backend provides the core APIs and services required for authentication, transportation data management, journey searching, candidate route generation, and route-related processing.

This report documents the testing activities performed on the BestRoute backend APIs to verify that the implemented functionality operates according to the expected requirements.

2. Testing Objectives

The main objectives of backend testing are:

- Verify that the backend APIs function according to their expected behavior.
- Verify that valid requests are processed successfully.
- Verify that invalid requests are handled appropriately.
- Verify that transportation data can be created, retrieved, updated, and deleted correctly.
- Verify that journey search returns appropriate transportation results.
- Verify that candidate routes can be generated from available transportation data.
- Verify that the system handles situations where no suitable route is available.
- Identify and document any errors or unexpected behavior during testing.

3. Testing Environment

Component	Tool / Technology
Backend Runtime	Node.js
Backend Framework	Express.js
Database	MongoDB
API Testing	Postman
Code Editor	Visual Studio Code
Version Control	Git / GitHub

4. Testing Scope

The backend testing covers the main API functionality implemented for the BestRoute system.

The testing scope includes:

- User authentication APIs
- Transportation data APIs
- Transportation CRUD operations
- Journey Search API
- Candidate Route Generation
- No Route Available scenario

The testing focuses on verifying API responses, request handling, data processing, and expected system behavior.

5. Test Cases

5.1 Authentication Testing

Test ID	Test Case	Expected Result	Actual Result	Status
AUTH-001	User registration with valid data	User should be registered successfully	User was registered successfully and the registration process completed without errors.	PASS
AUTH-002	Login with valid credentials	User should be logged in successfully	User logged in successfully using the valid credentials and was redirected to the appropriate page.	PASS
AUTH-003	Login with invalid credentials	Appropriate error should be returned	Login failed as expected and an appropriate error message was displayed for the invalid credentials.	PASS

5.2 Transportation API Testing

Test ID	Test Case	Expected Result	Actual Result	Status
TRANS-001	Create transportation record	Record should be created	Transportation record was created successfully and the created record was returned by the API.	PASS
TRANS-002	Get transportation records	Transportation records should be returned	Transportation records were retrieved successfully and the available records were returned by the API.	PASS
TRANS-003	Get transportation by ID	Correct record should be returned	The transportation record corresponding to the specified ID was retrieved successfully.	PASS
TRANS-004	Update transportation record	Record should be updated	The transportation record was updated successfully and the updated record was returned by the API.	PASS
TRANS-005	Delete transportation record	Record should be deleted	The transportation record was deleted successfully and the API confirmed the deletion.	PASS

5.3 CRUD Testing

Operation	Test ID	Expected Result	Status
Create	CRUD-001	Data successfully created	Data was created successfully and the newly created record was returned.
Read	CRUD-002	Data successfully retrieved	Data was retrieved successfully and the requested record was returned.
Update	CRUD-003	Data successfully updated	Data was updated successfully and the updated record was returned.
Delete	CRUD-004	Data successfully deleted	Data was deleted successfully and the deletion was confirmed by the API.

5.4 Journey Search API Testing

Test ID	Test Case	Expected Result	Actual Result	Status
SEARCH-001	Search journey with valid origin and destination	Matching transportation options should be returned	Matching transportation options were successfully retrieved for the specified origin and destination.	PASS
SEARCH-002	Search with unavailable location/criteria	Appropriate response should be returned	The API returned an appropriate response indicating that no matching transportation options were available for the specified criteria.	PASS

5.5 Candidate Route Generation Testing

Test ID	Test Case	Expected Result	Actual Result	Status
ROUTE-001	Generate candidate routes for valid journey	Candidate routes should be generated	Candidate routes were successfully generated for the specified journey.	PASS
ROUTE-002	Generate routes with available transportation combinations	Suitable route combinations should be returned	Suitable route combinations were successfully generated and returned based on the available transportation options.	PASS

5.6 No Route Scenario Testing

Test ID	Test Case	Expected Result	Actual Result	Status
ROUTE-003	Search for a journey where no suitable route exists	System should return an appropriate no-route response	No route available response was returned successfully	PASS

6. Test Results Summary

Testing Area	Total Tests	Passed	Failed
Authentication	3	3	0
Transportation API	5	5	0
CRUD	4	4	0
Journey Search	2	2	0
Candidate Route Generation	2	2	0
No Route Scenario	1	1	0
Total	17	17	0

7. Defects and Issues

During the testing process, no critical defects or functional issues were identified. All 17 test cases were executed successfully, and the actual results matched the expected results.

- No defects were identified in the authentication functionality.
- Transportation API operations including create, read, update, and delete were completed successfully.
- CRUD operations worked as expected.
- Journey search returned the expected transportation options and appropriate responses for unavailable criteria.
- Candidate route generation successfully produced suitable route combinations.
- The no-route scenario returned the expected response when no suitable route was available.
- No failed test cases were recorded during the testing process.

8. Conclusion

The testing process was successfully completed for the transportation and journey routing functionalities. A total of 17 test cases were executed covering authentication, transportation API operations, CRUD operations, journey search, candidate route generation, and no-route scenarios.

All 17 test cases passed successfully, resulting in a 100% pass rate. The actual results matched the expected results for all tested functionalities, and no critical defects or issues were identified during testing.

Based on the completed test cases, the tested functionalities operated as expected under the defined test conditions.